SCOB032 COMPUTER NETWORKS
Master Examination Study Guide

VERSION 3 — MASTER COMPLETE EDITION

Term Test 1 • Term Test 2 / Special Test 2 • Final Examination
University of Limpopo • Dr R.L. Moila

Full Version-1 teaching text restored • Original textbook figures embedded • Missing past-paper diagrams (Mobile IP triangle, Asymmetric crypto) added • FLSM/Classful subnetting and MANET/DSR sections added • TCP handshake arithmetic verified correct

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1. Source & Scope Note

This guide is constructed strictly from the official SCOB032 Master Exam & Tests Prep Guide (assessment scopes for Term Test 1, Term Test 2/Special Test 2 and the Final Examination), the prescribed textbooks Computer Networks (Tanenbaum & Wetherall, 5th edition) and Data Communications and Networking (Forouzan, 5th edition), and the complete set of lecture slide decks supplied for the module. Every topic that appears in the official scopes is treated. Supplementary explanations are clearly distinguished from source claims. Page and figure references are given where they can be verified.

Examination priority principle: Topics that appear repeatedly in past papers (manual VLSM, Hidden Terminal + RTS/CTS, OSI Layer 1–4 troubleshooting, TCP 3-way handshake arithmetic, Mobile IP direct vs indirect routing, transmission media selection, switching techniques, and security/CIA triad) receive the deepest treatment and the largest number of worked examples.

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3. Examination Strategy

The assessments reward precise technical language, correct layer identification, accurate arithmetic (especially VLSM and Sequence/ACK numbers), and the ability to evaluate flawed technician proposals. Marks are lost for vague answers, incorrect layer attribution, and arithmetic slips.

EXAMINATION TIP

Always state the OSI or TCP/IP layer first when asked to troubleshoot. Then give one concrete function of that layer that relates directly to the symptom. Finally explain why a lower or higher layer cannot solve the problem alone.

Cognitive levels tested

- Recall & define (topologies, ports, criteria, CIA triad)
- Explain & differentiate (CSMA/CD vs CSMA/CA, circuit vs packet switching, parity vs Hamming, direct vs indirect Mobile IP)
- Calculate & demonstrate (VLSM allocations, Sequence/ACK numbers, subnet metrics)
- Evaluate technician traps (broadcast address assignment, FLSM waste, wrong medium choice)
- Apply to scenarios (media selection over mountains, departmental segmentation, handoff)

4. Network Fundamentals, Types, Topologies & System Characteristics

4.1 What a computer network is

A computer network is a group of devices (hosts/nodes) that can communicate with each other over links. Each host has a unique address. Data communication is the movement of computer information from one point to another by means of electrical or optical transmission systems. Telecommunications is broader and includes voice and video.

4.2 Types of networks by geographic scope

LAN (Local Area Network) – connects devices in a home, office or campus building. High speed, limited geographic range, relatively low cost, easier security management. Single point of failure risk if the central device fails.

MAN (Metropolitan Area Network) – spans a city or metropolitan area; typically interconnects multiple LANs. Higher bandwidth than WAN for urban distances, but higher infrastructure cost.

WAN (Wide Area Network) – interconnects networks across cities, countries or continents (the Internet is the largest example). Greater latency and cost; relies on third-party providers.

4.3 Intranet vs Extranet

An **intranet** uses Internet technologies to share information inside an organisation and is open only to authorised internal users (employees accessing budgets, calendars, payroll). An **extranet** extends Internet technologies to selected external partners (suppliers, customers) so they can access authorised resources (inventory, order status). Extranets improve B2B coordination but introduce design challenges: authentication of external users, secure channels, and policy control of what each partner may see.

EXAMINATION TIP – EXTRANET

Past papers ask for benefits (improved partner coordination, faster information exchange) and challenges (security of external access, complexity of design and implementation, need for clear access policies). Always give a concrete example such as a bank giving selected partners controlled access to account or order information.

4.4 System upgrade characteristics & network criteria

When upgrading an online learning platform (or any networked system) four fundamental characteristics are considered:

- **Fault Tolerance** – the system continues to operate (possibly with degraded performance) after a component fails.
- **Scalability** – the system can grow (more users, more traffic, more sites) without a complete redesign.
- **Quality of Service (QoS)** – the ability to give different applications the performance they need (low delay for voice/video, high throughput for bulk transfer).
- **Security** – protection of confidentiality, integrity and availability of data and services.

Three criteria for an effective and efficient network are repeatedly examined:

- **Performance** – throughput, delay, jitter, response time.
- **Reliability** – availability, frequency of failure, recovery time, robustness.
- **Security** – protection against unauthorised access, modification and denial of service.

4.5 Network topologies

Topology describes the arrangement of nodes and connecting links.

Bus topology

All devices share a single backbone cable. Transmission is typically unidirectional. Advantages: inexpensive, easy to understand, simple to extend by joining cables. Disadvantages: the central cable is a single point of failure; if it breaks the whole network fails; performance degrades with heavy traffic; limited cable length. A physical bus often implements a logical bus. When the central cable fails, all communication stops. For rewiring an office that previously used a bus, a star topology is the preferred modern replacement.

Star topology

Every device has a dedicated link to a central hub or switch. Advantages: failure of one link or device does not bring down the rest of the network; easy to add or remove devices; straightforward troubleshooting. Disadvantages: the central device is a single point of failure; higher cabling cost. Star topology with Layer-2 switches is the recommended choice for departmental segmentation (HR, IT, Marketing) because each department's traffic can be isolated and performance is improved by MAC learning instead of flooding.

Ring topology

Each device is connected to exactly two neighbours, forming a closed loop. Data travels in one direction (or two directions in a dual ring). A token-passing discipline is often used. Advantages: deterministic access under load. Disadvantages: a single break can disrupt the ring (unless dual-ring protection is present); adding or removing a node temporarily breaks the ring.

Mesh topology

Every node is connected to every other node (full mesh) or to a subset (partial mesh). Number of links in a full mesh of n nodes is n(n–1)/2. Advantages: high robustness and privacy; alternative paths exist if one link fails. Disadvantages: expensive cabling and complex configuration. Used where reliability is critical.

Tree and Hybrid

Tree is a hierarchical extension of bus and star. Hybrid combines two or more basic topologies and inherits their advantages and disadvantages.

KEY TAKEAWAY – TOPOLOGIES

For a small office with IT, HR and Marketing that need independent traffic, recommend Star topology implemented with Layer-2 switches. Explain that each department can be placed on its own switch port (or VLAN) so that broadcast and collision domains are separated and a cable failure affects only one device.

4.6 Original Textbook Figure — OSI Reference Model

Source: Tanenbaum & Wetherall, Computer Networks, 5th edition, Figure 1-20.

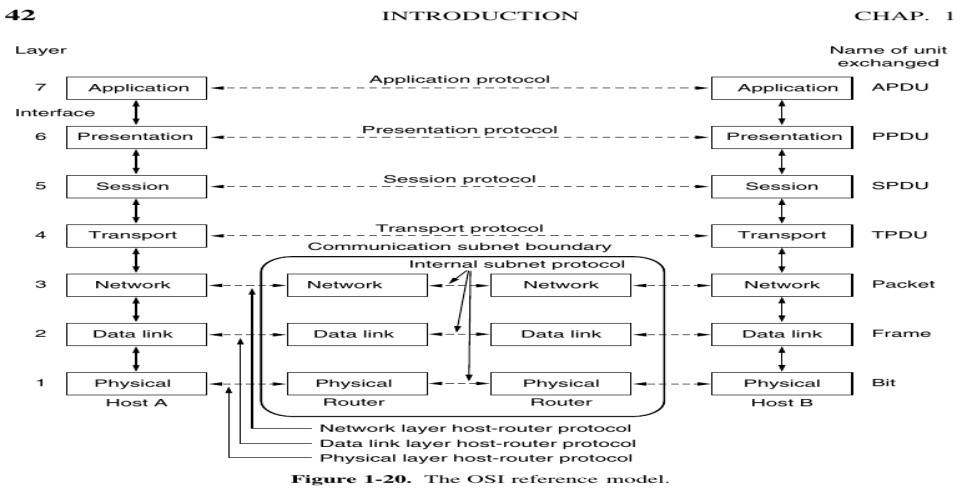


Figure 1-20. The OSI reference model.

4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

Below we will discuss each layer of the model in turn, starting at the bottom layer. Note that the OSI model itself is not a network architecture because it does not specify the exact services and protocols to be used in each layer. It just tells what each layer should do. However, ISO has also produced standards for all the layers, although these are not part of the reference model itself. Each one has been published as a separate international standard. The *model* (in part) is widely used although the associated protocols have been long forgotten.

Original textbook page containing Figure 1-20 — The OSI reference model (Tanenbaum 5e).

What it represents: The seven layers of the OSI reference model and the units exchanged at each layer (bits, frames, packets, TPDUs, etc.). Host A and Host B communicate peer-to-peer at each layer; intermediate routers operate only up to the Network layer.

Examination relevance: You must be able to identify which layer is responsible for a given symptom (weak signal → Layer 1; wrong application → Layer 4; cannot leave subnet → Layer 3; MAC delivery inside subnet → Layer 2).

5. Transmission Media, Impairments & Physical Layer

5.1 Guided vs unguided media

Guided (wired) media use a physical path: twisted-pair copper, coaxial cable, optical fibre. Unguided (wireless) media use electromagnetic waves through free space: radio, microwave, infrared, satellite.

5.2 Twisted pair

Two insulated copper wires twisted together to reduce electromagnetic interference (crosstalk). Cheap, flexible, easy to install. Limited distance and bandwidth compared with fibre; susceptible to EMI and lightning. Widely used in LANs (Ethernet) and telephone local loops.

5.3 Coaxial cable

Central conductor, insulating dielectric, metallic shield, outer jacket. Better immunity to interference and higher bandwidth than twisted pair; used for cable television, some older Ethernet, and longer runs. Thicker and less flexible than twisted pair.

5.4 Optical fibre

Glass or plastic core surrounded by cladding of lower refractive index; light is guided by total internal reflection. Extremely high bandwidth, very low attenuation, immune to EMI and lightning, small size and light weight. Preferred for long-distance and high-speed backbone links and for environments with electrical noise or thunderstorm risk. Drawbacks: higher installation cost, fragility, specialised splicing equipment.

5.5 Microwave and satellite

Microwave is line-of-sight terrestrial radio. Requires a clear path; tall buildings and mountains block the signal. Relay towers are placed every 20–50 km. Satellite provides wide coverage via uplink/downlink but introduces propagation delay and is affected by heavy rain (rain fade).

5.6 Transmission impairments

• **Attenuation** – gradual loss of signal strength with distance. Overcome by repeaters (digital) or amplifiers (analogue).
• **Distortion** – different frequency components travel at different speeds and arrive with altered shape.
• **Noise** – unwanted energy (thermal, impulse, crosstalk, intermodulation, EMI/RFI) that corrupts the signal. Lightning and thunderstorms are classic sources of impulse noise on copper media.

EXAMINATION TIP – MEDIA SELECTION

Scenario: two offices 25 km apart across tall buildings and mountainous terrain. Microwave is unsuitable because it requires line-of-sight; mountains and buildings block the path. Recommend optical fibre: high bandwidth, low attenuation over distance, immunity to EMI and lightning. If cable installation is impossible, a multi-hop microwave or satellite solution with intermediate relays can be considered, but fibre remains preferred when feasible.

5.7 Multiplexing (brief)

FDM – frequency division; channels occupy different frequency bands. WDM – optical analogue of FDM using different wavelengths of light on one fibre. TDM – time division; channels take turns in time slots (T1 is a classic example). Multiplexing allows many conversations or data streams to share one physical medium.

6. Switching Techniques & Connecting Devices

6.1 Circuit, Message and Packet switching

Circuit switching – a dedicated end-to-end path is established before data transfer and held for the duration of the session. Guaranteed bandwidth and constant delay; inefficient if the path is idle. Classic telephone network. Suitable for real-time voice.

Message switching – the entire message is stored at each intermediate node and forwarded when the next link is free (store-and-forward). No dedicated path; good channel utilisation; unsuitable for real-time traffic because of store-and-forward delay.

Packet switching – the message is divided into packets that travel independently (possibly over different routes) and are reassembled at the destination. Efficient for bursty data; robust; the dominant technique on the Internet. Two variants: virtual-circuit packet switching and datagram packet switching.

TECHNICIAN TRAP – SWITCHING

A technician argues that message switching should be used for real-time voice because intermediate nodes store and forward the complete message. This is incorrect: the store-and-forward delay makes message switching unsuitable for real-time voice. Circuit switching (or packet switching with strong QoS) is appropriate for real-time voice; packet switching is appropriate for overnight bulk file transfer.

6.2 Connecting devices by layer

• **Passive hub** – simple connector; operates below the physical layer; collision domain is the whole shared medium.
• **Repeater / Active hub** – physical layer; regenerates the bit stream; extends distance but does not filter; still one collision domain.
• **Bridge / Layer-2 switch** – physical + data-link; learns MAC addresses and forwards frames only to the correct port; each port is a separate collision domain; still one broadcast domain unless VLANs are used.
• **Router / Layer-3 switch** – physical + data-link + network; forwards packets on the basis of IP addresses; separates broadcast domains; performs inter-network routing.
• **Gateway** – can operate at all layers; performs protocol translation between different architectures.

Installing additional repeaters does not solve performance problems caused by a large collision domain: every frame is still flooded to every segment. Replacing hubs with Layer-2 switches creates separate collision domains and uses MAC learning, which is the correct solution for a 40-computer LAN suffering from collisions. A Layer-2 switch alone cannot route between two different IP subnets; a router (or Layer-3 switch) is required.

7. Data-Link Layer: Framing, Error Control & MAC

7.1 Core responsibilities

The data-link layer moves frames from one node to an adjacent node over a single link. It performs framing, physical addressing (MAC), error detection/correction, flow control and medium access control on shared media.

7.2 Error detection vs error correction

Parity – one extra bit makes the number of 1s even (even parity) or odd (odd parity). Simple and fast; detects single-bit errors; fails on even numbers of bit flips.

Checksum – arithmetic sum of data words; simple; used in IP/TCP/UDP headers.

CRC (Cyclic Redundancy Check) – treats the frame as a polynomial and divides by a generator polynomial; the remainder is appended. Extremely good at detecting burst errors; used in Ethernet, Wi-Fi, etc.

Hamming code – adds multiple parity bits at power-of-two positions so that the pattern of parity failures identifies the exact bit that is wrong; therefore it can *correct* single-bit errors (forward error correction). Parity only detects; Hamming (and more powerful FEC codes) can correct without retransmission.

ARQ (Automatic Repeat reQuest) – receiver requests retransmission of corrupted frames (Stop-and-Wait, Go-Back-N, Selective Repeat). FEC adds redundancy so the receiver can repair errors without a reverse channel; used when retransmission is costly (satellite, real-time streams).

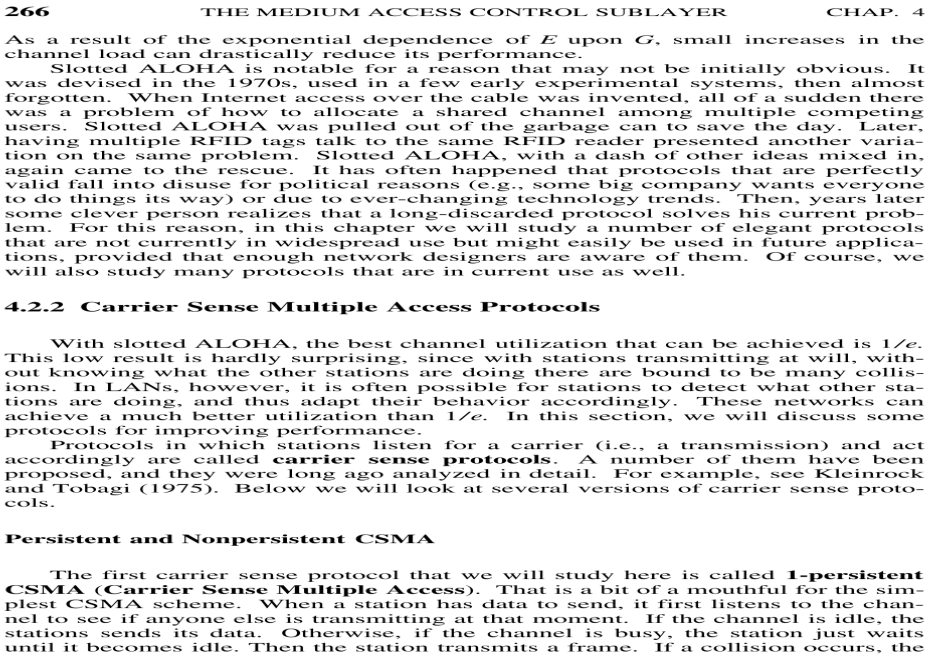
8. Medium Access Protocols

8.1 Contention vs controlled access

Contention (random access) – stations transmit when the medium appears free; collisions are possible (ALOHA, CSMA variants). Controlled access – a station transmits only when authorised (polling, token passing, reservation). Under light load contention has lower overhead; under heavy load controlled access is more efficient.

8.2 Pure ALOHA and Slottted ALOHA

Pure ALOHA – a station transmits whenever it has a frame; if a collision occurs it waits a random time and retries. Vulnerable period is two frame times; maximum throughput approximately 18 %. Slotted ALOHA – time is divided into slots; a station may transmit only at the beginning of a slot; vulnerable period is one frame time; maximum throughput approximately 37 %. Both are simple but inefficient under load.



266 THE MEDIUM ACCESS CONTROL SUBLAYER CHAP. 4

As a result of the exponential dependence of E upon G, small increases in the channel load can drastically reduce its performance.

Slotted ALOHA is notable for a reason that may not be initially obvious. It was devised in the 1970s, used in a few early experimental systems, then almost forgotten. When Internet access over the cable was invented, all of a sudden there was a problem of how to allocate a shared channel among multiple competing users. Slotted ALOHA was pulled out of the garbage can to save the day. Later, having multiple RFID tags talk to the same RFID reader presented another variation on the same problem. Slotted ALOHA, with a dash of other ideas mixed in, again came to the rescue. It has often happened that protocols that are perfectly valid fall into disuse for political reasons (e.g., some big company wants everyone to do things its way) or due to ever-changing technology trends. Then, years later some clever person realizes that a long-discarded protocol solves his current problem. For this reason, in this chapter we will study a number of elegant protocols that are not currently in widespread use but might easily be used in future applications, provided that enough network designers are aware of them. Of course, we will also study many protocols that are in current use as well.

4.2.2 Carrier Sense Multiple Access Protocols

With slotted ALOHA, the best channel utilization that can be achieved is $1/e$. This low result is hardly surprising, since with stations transmitting at will, without knowing what the other stations are doing there are bound to be many collisions. In LANs, however, it is often possible for stations to detect what other stations are doing, and thus adapt their behavior accordingly. These networks can achieve a much better utilization than $1/e$. In this section, we will discuss some protocols for improving performance.

Protocols in which stations listen for a carrier (i.e., a transmission) and act accordingly are called **carrier sense protocols**. A number of them have been proposed, and they were long ago analyzed in detail. For example, see Kleinrock and Tobagi (1975). Below we will look at several versions of carrier sense protocols.

Persistent and Nonpersistent CSMA

The first carrier sense protocol that we will study here is called **1-persistent CSMA (Carrier Sense Multiple Access)**. That is a bit of a mouthful for the simplest CSMA scheme. When a station has data to send, it first listens to the channel to see if anyone else is transmitting at that moment. If the channel is idle, the stations sends its data. Otherwise, if the channel is busy, the station just waits until it becomes idle. Then the station transmits a frame. If a collision occurs, the

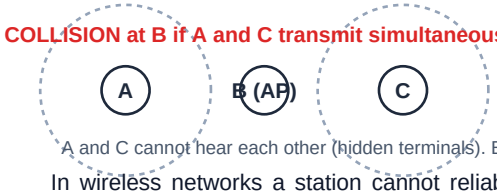
Textbook page from Tanenbaum Ch. 4 (ALOHA region) — original source material.

8.3 CSMA/CD (wired Ethernet)

Carrier Sense Multiple Access with Collision Detection. A station listens before transmitting (carrier sense). If the medium is idle it transmits. While transmitting it continues to listen; if it detects a collision it aborts, sends a jam signal, and retries after a random exponential back-off. Used on classic half-duplex Ethernet (shared coaxial or hubs). Modern switched full-duplex Ethernet does not need CSMA/CD because each link is a dedicated collision-free channel.

8.4 CSMA/CA and the Hidden Terminal Problem

SIMPLIFIED TEACHING DIAGRAM — BASED ON TEXTBOOK / LECTURE MATERIAL



A and C cannot hear each other (hidden terminals). Both reach B.

In wireless networks a station cannot reliably listen while transmitting (its own signal drowns out others) and the hidden-terminal problem exists: stations A and C can both communicate with B but cannot hear each other. Consequently A and C may transmit simultaneously, colliding at B. CSMA/CD is therefore inappropriate for wireless media.

CSMA/CA (Collision Avoidance) is used instead. A station senses the medium; if busy it defers. Even if idle it may wait a random back-off. Optional RTS/CTS handshake: sender transmits a short Request-to-Send; the access point (or receiver) replies with

Clear-to-Send; every station that hears the CTS remains silent for the duration of the subsequent data exchange. This silences the hidden terminals and greatly reduces collisions.

STUDENT TRAP – CSMA/CD ON WIRELESS

A student proposes replacing CSMA/CA with CSMA/CD on a Wi-Fi network, arguing that devices can simply detect collisions after they occur. This is inappropriate: (1) a wireless transmitter cannot hear weak signals from hidden terminals while it is transmitting; (2) the hidden-terminal problem means collisions occur at the receiver even though the senders never detect them. CSMA/CA with RTS/CTS is the correct approach.

9. Network Layer: IPv4 Addressing, FLSM & Manual VLSM

9.1 IPv4 address structure

An IPv4 address is 32 bits, usually written in dotted-decimal (e.g. 192.168.10.25). It contains a network portion and a host portion. The subnet mask (or prefix length /n) indicates how many bits belong to the network. Classful addressing (A/B/C) is obsolete but still appears in exam questions for historical classification. Modern practice is classless (CIDR).

9.2 FLSM vs VLSM + Class B/C worked example

Fixed-Length Subnet Masking (FLSM) – every subnet uses the same mask. Simple but wastes addresses when subnets need very different numbers of hosts. Variable-Length Subnet Masking (VLSM) – different subnets may use different masks. Addresses are allocated from largest host requirement to smallest so that each block fits tightly. VLSM is required in the SCOB032 examinations and calculators are not allowed.

FLSM / Class B worked example (borrowing bits)

Network 172.16.0.0 (Class B, default /16). Requirement: 8 equal subnets.
Bits to borrow: $2^3 = 8 \rightarrow$ borrow 3 bits $\rightarrow$ new prefix /19.
Subnet mask: 255.255.224.0. Block size = 256 – 224 = 32 (in the third octet).
Usable hosts per subnet = $2^{13} - 2 = 8190$.
Subnet 0: 172.16.0.0 – 172.16.31.255
Subnet 1: 172.16.32.0 – 172.16.63.255
Subnet 2: 172.16.64.0 – 172.16.95.255
Subnet 3: 172.16.96.0 – 172.16.127.255 ← third subnet
First usable of Subnet 3: 172.16.96.1 ; Last usable: 172.16.127.254 ; Broadcast: 172.16.127.255.

Block-size shortcut (find network of a host)

Host 200.45.34.56/28. Mask 255.255.255.240 $\rightarrow$ block size = 256 – 240 = 16.
Networks: 0, 16, 32, 48, … 56 falls in the block starting at 48.
Network address = 200.45.34.48 ; Broadcast = 200.45.34.63 ; First usable = 200.45.34.49 ; Last usable = 200.45.34.62.

9.3 Essential subnet metrics

For any subnet you must be able to state:
• CIDR prefix length and dotted-decimal mask
• Network (subnet) address – all host bits 0
• First usable host address – network + 1
• Last usable host address – broadcast – 1
• Broadcast address – all host bits 1
• Number of usable hosts = $2^h - 2$ (where h = host bits)

9.4 Worked VLSM example (matching past-paper style)

Problem (enterprise style)

Address space: 10.10.0.0/16. Requirements (hosts): LAN1 300, LAN3 100, LAN4 55, LAN2 30, WAN1 2. Allocate in descending order of host count using VLSM. List Network, First usable, Last usable, Broadcast for each. State the first unallocated address after the allocations.

Solution

1. Host bits needed (choose smallest power of 2 that satisfies usable hosts = 2^h–2):
• 300 hosts $\rightarrow 2^9$–2 = 510 $\rightarrow$ /23
• 100 hosts $\rightarrow 2^7$–2 = 126 $\rightarrow$ /25
• 55 hosts $\rightarrow 2^6$–2 = 62 $\rightarrow$ /26
• 30 hosts $\rightarrow 2^5$–2 = 30 $\rightarrow$ /27
• 2 hosts $\rightarrow 2^2$–2 = 2 $\rightarrow$ /30

Subnet	Network	First usable	Last usable	Broadcast
LAN1 /23	10.10.0.0	10.10.0.1	10.10.1.254	10.10.1.255
LAN3 /25	10.10.2.0	10.10.2.1	10.10.2.126	10.10.2.127
LAN4 /26	10.10.2.128	10.10.2.129	10.10.2.190	10.10.2.191
LAN2 /27	10.10.2.192	10.10.2.193	10.10.2.222	10.10.2.223
WAN1 /30	10.10.2.224	10.10.2.225	10.10.2.226	10.10.2.227

First unallocated address: 10.10.2.228

Technician trap: a technician assigns 10.10.2.150 to a host in LAN3. 10.10.2.150 lies inside 10.10.2.128/26 (LAN4), not inside 10.10.2.0/25 (LAN3). The assignment is incorrect.

9.5 Second worked example (172.20.0.0/23 style)

Parent block 172.20.0.0/23 (512 addresses). Requirements: Administration 120, Finance 60, Research 28, Security 12. Descending VLSM:

- Admin 120 → /25 (126 usable) → 172.20.0.0/25, Broadcast 172.20.0.127
- Finance 60 → /26 (62 usable) → 172.20.0.128/26, Broadcast 172.20.0.191
- Research 28 → /27 (30 usable) → 172.20.0.192/27, Broadcast 172.20.0.223
- Security 12 → /28 (14 usable) → 172.20.0.224/28, Broadcast 172.20.0.239

First unallocated: 172.20.0.240

COMMON MISTAKE – VLSM

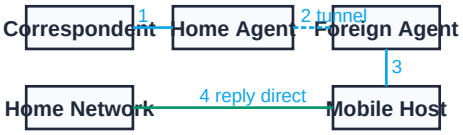
Never assign a broadcast address or network address to a host. Always allocate from largest to smallest host requirement so that later smaller blocks fit into the remaining space. Write the full four metrics (Network, First, Last, Broadcast) for every subnet; examiners allocate marks for each.

10. Routing, Mobile IP, MANET/DSR & Congestion Basics

10.1 Direct vs Indirect routing (Mobile IP)

A mobile host has a permanent home address and, when away, a temporary care-of address on the visited network. The home agent resides on the home network; a foreign agent may reside on the visited network (or the mobile may act as its own foreign agent).

SIMPLIFIED TEACHING DIAGRAM — INDIRECT (TRIANGLE) ROUTING



INDIRECT (triangle): CN->HA->FA->MH ; MH replies direct to CN

Indirect routing (triangle routing)

Correspondent sends packets to the mobile's home address. Home agent intercepts them, tunnels them to the care-of address (foreign agent or mobile). Mobile replies directly to the correspondent. Path is correspondent → home agent → foreign agent → mobile (triangle). Transparent to the correspondent but adds latency, especially when correspondent and mobile are topologically close.

Direct routing

Correspondent first learns the care-of address (via home agent or binding update) and thereafter sends packets directly to the care-of address. More efficient path; the correspondent must be aware of mobility.

Handoff / hand-over – the process that transfers an active connection from one base station (or access point) to another as the mobile moves, preserving the ongoing session.

10.2 Infrastructure vs Ad-Hoc mode & MANET / DSR

Infrastructure mode – wireless hosts associate with a base station / access point that is connected to a wired network. The infrastructure provides addressing, routing and DNS. Handoff occurs when a mobile moves from one AP to another within the same subnet (IP address can stay the same).

Ad-hoc mode – no base station. Nodes communicate directly if within range and must self-organise routing for multi-hop paths. Used in emergency response, temporary networks and sensor networks.

MANET (Mobile Ad-hoc Network) – a self-configuring network of mobile devices connected by wireless links without fixed infrastructure.

Dynamic Source Routing (DSR) – an on-demand routing protocol for MANETs. When a node needs a route it floods a Route Request (RREQ). Intermediate nodes append their address and forward the RREQ. The destination (or a node with a cached route) replies with a Route Reply (RREP) that travels back along the reverse path, carrying the full source route. Subsequent data packets carry the complete route in the header. DSR uses route caching to reduce future floods. It is reactive (routes discovered only when needed) and source-routed.

11. Transport Layer: TCP, UDP & Handshake Math

11.1 TCP vs UDP

TCP – connection-oriented, reliable, ordered, flow-controlled and congestion-controlled. Uses sequence numbers, acknowledgements, retransmissions and a sliding window. Suitable for file transfer, web pages, email. Overhead is higher.

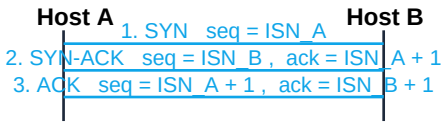
UDP – connectionless, unreliable, unordered, no flow or congestion control. Minimal overhead. Suitable for DNS queries, real-time audio/video, applications that implement their own reliability.

11.2 Standard ports (must memorise)

HTTP 80, HTTPS 443, SSH 22, SMTP 25, DNS 53 (UDP/TCP), FTP 21 (control), FTP data 20.

11.3 TCP three-way handshake – arithmetic (CORRECTED)

SIMPLIFIED TEACHING DIAGRAM — BASED ON TEXTBOOK (Tanenbaum Ch. 6) AND LECTURE SLIDES



Host A chooses an initial sequence number ISN_A. Host B chooses ISN_B. The handshake is:

1. A → B : SYN, Sequence = ISN_A
2. B → A : SYN-ACK, Sequence = ISN_B, Acknowledgement = $ISN_A + 1$
3. A → B : ACK, Sequence = $ISN_A + 1$, Acknowledgement = $ISN_B + 1$

Worked example (verified correct)

$ISN_A = 1543$, $ISN_B = 2780$.

- B's SYN-ACK carries Acknowledgement = 1543 + 1 = **1544**.
- A's final ACK carries Sequence = 1544 and Acknowledgement = 2780 + 1 = **2781**.

Second example: ISN_Student = 4250, ISN_Server = 7600.

- SYN-ACK: Sequence = 7600, ACK = 4251
- Final ACK: Sequence = 4251, ACK = 7601

Always draw the three arrows, label the flags, and write both Sequence and Acknowledgement numbers on each segment. Examiners allocate separate marks for the diagram and for each correct number.

12. Application Layer Services & Security Foundations

12.1 DNS and DHCP

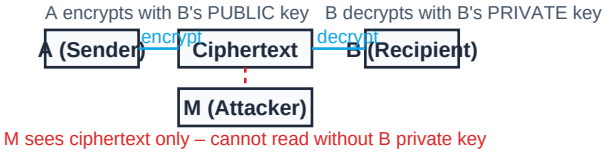
DNS resolves human-readable names to IP addresses (iterative or recursive queries; UDP port 53). DHCP dynamically assigns IP addresses and related parameters (DORA: Discover, Offer, Request, Acknowledge).

12.2 CIA Triad

- **Confidentiality** – information is accessible only to authorised parties (encryption, access control).
- **Integrity** – information is not altered in an unauthorised manner (hashes, digital signatures, MACs).
- **Availability** – authorised users can access resources when needed (redundancy, DoS protection, backups).

12.3 Asymmetric (public-key) cryptography

SIMPLIFIED TEACHING DIAGRAM — BASED ON LECTURE / PAST-PAPER REQUIREMENT (A, M, B)



Each party has a public key (freely distributed) and a private key (kept secret). To send a confidential message to B, A encrypts with B's public key; only B can decrypt with B's private key. An attacker M who intercepts the ciphertext cannot read it without B's private key. Digital signatures are created with the sender's private key and verified with the sender's public key.

12.4 Passive vs Active attacks & DoS

Passive attacks (eavesdropping, traffic analysis) aim to obtain information without altering it; they are hard to detect but can be prevented by encryption. Active attacks (masquerade, replay, message modification, denial of service) alter or disrupt the system; they are easier to detect but harder to prevent completely. A DoS/DDoS attack targets Availability by overwhelming a resource (e.g., flooding a registrar database). Mitigations include rate limiting, firewalls, redundant capacity and traffic filtering.

13. Selected High-Yield Worked Questions

Q1 – OSI Troubleshooting (typical 10-mark scenario)

Computer A can communicate locally but cannot reach outside networks; Computer B receives signals that are too weak; Computer C receives data correctly but it is delivered to the wrong application.

Model answer outline

A → Layer 3 (Network). Function: logical addressing / routing / default gateway. The host can reach the local subnet (Layer 2 works) but cannot leave it, pointing to missing or incorrect default gateway or routing failure.

B → Layer 1 (Physical). Function: bit transmission / signal strength. Weak signal is attenuation or a faulty cable/connector/transceiver.

C → Layer 4 (Transport). Function: port numbers / process-to-process delivery. Data arrives at the correct host but is handed to the wrong process, indicating a port-mapping error.

Troubleshooting only the Data-Link layer cannot solve all three problems because the symptoms involve Layers 1, 3 and 4; Data-Link (MAC delivery inside a single subnet) is already working for A and C and is irrelevant to B's signal-strength problem.

Q2 – TCP Handshake Arithmetic

ISN_Student = 4250, ISN_Server = 7600. Give Sequence and ACK numbers for the SYN-ACK and the final ACK.

SYN-ACK (Server → Student): Sequence = 7600, ACK = 4251
Final ACK (Student → Server): Sequence = 4251, ACK = 7601

Q3 – VLSM Technician Trap

After a correct VLSM allocation, a technician assigns an address that falls inside a different subnet's range. The answer must recalculate (or quote) the range of the claimed subnet and show that the address lies outside it, therefore the assignment is invalid.

14. Quick-Reference Sheet

Ports: 20/21 FTP, 22 SSH, 25 SMTP, 53 DNS, 80 HTTP, 443 HTTPS

Usable hosts: $2^h - 2$

TCP Handshake:
SYN (seq = ISN_A)
SYN-ACK (seq = ISN_B, ack = ISN_A+1)
ACK (seq = ISN_A+1, ack = ISN_B+1)

OSI quick map:
1 Physical – bits, cables, signals, attenuation
2 Data-Link – frames, MAC, switches, error detection
3 Network – packets, IP, routers, subnetting
4 Transport – segments, TCP/UDP, ports, handshake
5–7 Session/Presentation/Application – dialog, syntax, user services

CIA: Confidentiality, Integrity, Availability

Mobile IP: Home address + Care-of address; Home Agent tunnels to Foreign Agent (indirect) or correspondent sends direct after learning CoA.

Block size: 256 – last mask octet (for /24 and longer).

15. Revision Checklist

- ■ I can list the four system-upgrade characteristics and the three network criteria.
- ■ I can draw and compare Bus, Star, Ring and Mesh; I know when to recommend Star + L2 switches.
- ■ I can select the correct medium for a mountainous 25 km link and justify with three characteristics.
- ■ I can explain why extra repeaters do not fix a hub-based collision domain.
- ■ I can distinguish circuit, message and packet switching and choose the right one for voice vs bulk transfer.
- ■ I can explain parity vs Hamming and when each is appropriate.
- ■ I can describe CSMA/CD, CSMA/CA, the Hidden Terminal Problem and the RTS/CTS solution.
- ■ I can perform a complete VLSM allocation (descending hosts) and list Network / First / Last / Broadcast for every subnet.
- ■ I can perform FLSM / Class B or C subnetting by borrowing bits and using the block-size shortcut.
- ■ I can detect a technician trap that assigns a broadcast address or an address from the wrong subnet.
- ■ I can draw the TCP three-way handshake and compute Sequence and ACK numbers from given ISNs (ISN+1).
- ■ I can map common services to their standard ports.
- ■ I can explain direct vs indirect Mobile IP routing and the roles of Home Agent, Foreign Agent and Care-of Address.
- ■ I can describe Infrastructure vs Ad-hoc mode and how DSR uses RREQ/RREP.
- ■ I can define the CIA triad and give one control that protects each property.
- ■ I can distinguish passive from active attacks and recommend two mitigations for a DoS flood.
- ■ I can explain asymmetric encryption: encrypt with recipient public key, decrypt with recipient private key.

VERSION 3 — MASTER COMPLETE EDITION. Full Version-1 text restored. Textbook OSI figure and ALOHA-region page embedded. Hidden Terminal, TCP handshake, Mobile IP triangle and Asymmetric crypto diagrams added. FLSM/Classful subnetting and MANET/DSR sections added. TCP arithmetic verified: ack = ISN+1 and final ACK acknowledgement = 2781. Work every past-paper VLSM and handshake question under timed conditions. Good fortune in the examination.